



Material Compliance Guide

RoHS, REACH and DFARS: what they are, when they apply to our work, and how we prove it.

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The one-minute version

None of these rules apply to us directly. They apply to our customers, who pass them down on the PO. The trigger is always paper: a PO note, a drawing note, a supplier quality clause, or a declaration form. If the PO is silent, we do nothing extra and make no claim. If the PO calls one out, we source to it and cite it on the Certificate of Conformance.

Rule	What it restricts	Where it bites a sheet metal shop	How we prove it
RoHS EU Directive 2011/65/EU as amended by (EU) 2015/863 ("RoHS 3")	10 substances in electrical products: lead, mercury, cadmium, hex chrome, PBB, PBDE, 4 phthalates. Limit 0.1% (cadmium 0.01%).	Yellow zinc chromate and hex-chrome chem film (Alodine 1200). Leaded free-machining stock (12L14, 2011). Vinyl trim, rubber, foam.	Mill cert that says RoHS, or a distributor RoHS letter. Finisher cert stating trivalent. Hardware maker statement.
REACH EU Regulation (EC) 1907/2006, Article 33	Disclosure, not a ban. If any SVHC on the Candidate List is above 0.1% w/w, we must name it. List grows twice a year (240+ entries).	Same as RoHS plus: lead has NO alloying exemption under REACH, so 12L14 must be declared. Hex chrome baths. Cobalt and nickel compounds in some finishes.	Distributor RoHS/REACH letter (usually one combined letter). Finisher statement. Refresh yearly because the list changes.
DFARS 252.225-7009 Specialty Metals	Specialty metals in DoD deliverables must be melted in the USA or a qualifying country (NATO allies, Japan, Australia, Israel, others).	Stainless and alloy steel, titanium. Plain carbon steel and aluminum are NOT specialty metals, so foreign-melt A1008 and 5052 are fine.	Mill cert showing melt country. "DFARS domestic melt" written on our PO to the distributor for stainless and alloy steel.

Our stock at a glance

Material we buy	RoHS	REACH	DFARS specialty metal?	Notes
A1008 / A36 cold & hot rolled steel	Pass	Pass	No	Paperwork only. Need cert line or letter.
Galvanized / galvanneal (A653)	Pass	Pass	No	Zinc coating is fine. Any chromate passivation on the coil should be trivalent; ask if in doubt.
5052 / 6061 aluminum	Pass	Pass	No	Spec caps unlisted elements at 0.05% each, so lead cannot exceed the 0.1% limit.
304 / 316 stainless	Pass	Pass	YES	Foreign melt is a DFARS problem. Cert must show melt country.
4130 / alloy steel	Pass	Pass	YES	Same DFARS melt rule as stainless.
12L14 leaded steel	Pass (exempt)	DECLARE	No	Lead 0.15 to 0.35%. RoHS exemption 6(a) covers it. REACH requires naming lead on the declaration.
2011 aluminum	RISK	DECLARE	No	Lead 0.2 to 0.6% can exceed the 0.4% RoHS exemption limit. Avoid on RoHS jobs; use 6061.
360 brass	Pass (exempt)	DECLARE	No	Lead ~3%. RoHS exemption 6(c) allows up to 4%. REACH declaration required.

THE PATTERN

Bare sheet almost never fails RoHS or REACH. The outside finish and the hardware are where parts actually fail. Stainless is where DFARS fails. Get the letters on file once and every flowdown becomes a paperwork step, not a sourcing scramble.

RoHS in detail

Restriction of Hazardous Substances. A European Union product law for electrical and electronic equipment. RoHS 1 (2002/95/EC) listed six substances. RoHS 2 (2011/65/EU, 2013) widened the product scope and added CE-marking duties. RoHS 3 is the informal name for amendment (EU) 2015/863, which added four phthalates from July 2019. When a customer writes RoHS, RoHS 2, or RoHS 3 today they all mean the same ten-substance list below.

Substance	Limit (w/w, per homogeneous material)	Where it shows up in our work
Lead (Pb)	0.1%	Alloying element in 12L14, 2011, 360 brass. Not in A1008, 5052, 6061, or stainless.
Cadmium (Cd)	0.01%	Cad plating. Trace in some old alloys. Not used on RoHS work.
Mercury (Hg)	0.1%	Not in metals. Ignore.
Hexavalent chromium (Cr VI)	0.1%	Yellow / olive zinc chromate, Alodine 1200 / 1201 chem film, some passivates. THE usual failure.
PBB, PBDE	0.1% each	Flame retardants in plastics. Only if we supply plastic parts.
DEHP, BBP, DBP, DIBP (phthalates)	0.1% each	Plasticizers in vinyl edge trim, PVC grommets, rubber bumpers, some foam tape.

Exemptions we rely on (Annex III)

- **6(a)** lead as an alloying element in steel, up to 0.35%. Covers 12L14.
- **6(b)** lead in aluminum, up to 0.4%. 2011 aluminum can exceed this. Use 6061 on RoHS jobs.
- **6(c)** lead in copper alloys, up to 4%. Covers 360 brass and most leaded bronze.

Exemptions carry expiry and renewal dates. If a customer is strict, confirm the current status before quoting a leaded alloy.

How to read a mill cert for RoHS

- **Best case:** the cert says it. Look in the Specification block for "RoHS compliant" next to the ASTM grade. Cite the cert number and heat on the CoC and you are done.
- **Chemistry check:** find Pb and Cd in the chemistry table. Lead under 0.1% and cadmium under 0.01% passes. Total chromium (Cr) on a steel cert is metallic chromium, not hexavalent; it does not count.
- **Not listed:** most steel and aluminum certs never report lead. For aluminum the spec's "others, each 0.05% max" proves it. For steel the spec has no lead cap, so the cert alone cannot prove compliance.
- **Then:** ask the distributor (TYRO) for their standing RoHS/REACH letter for that alloy family, or ask them to add "RoHS compliant" to the cert. Check for a page 2; remarks often land there.

Outside finishes: what to write on the PO to the finisher

Process	Fails RoHS	Passes RoHS	PO wording
Chem film (chromate conversion)	Alodine 1200 / 1201, MIL-DTL-5541 Type I	Alodine 5200, SurTec 650, MIL-DTL-5541 Type II (trivalent)	"MIL-DTL-5541 Type II, trivalent, RoHS compliant. State on cert."
Zinc plating	Yellow, olive drab, black hex chromate	Clear or blue trivalent passivate	"ASTM B633, trivalent clear, RoHS compliant. State on cert."
Anodize	Normally fine	Type II / III with nickel-acetate or hot-water seal	"RoHS compliant, state on cert." Dichromate seal is hex chrome; do not allow.
Passivation (stainless)	Normally fine	Citric or nitric per AMS 2700 / ASTM A967	"RoHS compliant, state on cert."
Powder coat	Normally fine	Nearly all modern powders	"RoHS compliant, state on cert." Ask for the powder TDS if pressed.

REMEMBER

Anodize, chem film, passivate, plating and powder coat are all OUTSIDE processes for us. The finisher's cert is what proves the finish. If their cert does not say trivalent or RoHS, the part is not proven, no matter what the mill cert says.

REACH in detail

Registration, Evaluation, Authorisation and Restriction of Chemicals. EU Regulation (EC) 1907/2006, in force since 2007. Where RoHS is a short fixed list with hard limits for electrical goods, REACH covers every chemical and every article sold in the EU. It works through the **SVHC Candidate List** (Substances of Very High Concern), which the European Chemicals Agency updates twice a year, typically January and June.

Our only obligation is **Article 33**: if any SVHC is present above 0.1% by weight of an article, tell the customer which one. Below that, or if none is present, say so. No registration, no fee, no testing. Naming a substance is compliance. Hiding it is the violation.

SVHCs that touch sheet metal work

Substance	Source in our work	What we do
Lead (listed 2018)	12L14, 2011 aluminum, leaded brass. No alloying exemption under REACH.	Declare it by name on the form or CoC. Offer 1215 / 6061 / lead-free brass if the customer objects.
Chromium trioxide and other Cr(VI) compounds	Hex chrome chem film and yellow chromate baths. Also on the Authorisation list, which is why EU platers are dropping it.	Specify trivalent. Finisher states it on cert.
Cobalt and nickel compounds	Some passivation and plating chemistries. Usually below threshold in the finished part.	Ask the finisher to confirm on their letter.
Phthalates (DEHP, DBP, BBP, DIBP)	Vinyl trim, PVC grommets, rubber bumpers, foam tape.	Buy from suppliers with a REACH statement, or leave the plastic parts to the customer.
Cadmium and its compounds	Cad plating.	Not used on REACH or RoHS work.

Who asks for REACH

- **Aerospace and defense** customers ask for REACH more than RoHS, because RoHS exempts their equipment and REACH does not.
- **Medical** customers usually ask for both.
- **Electronics** customers ask for RoHS first and REACH as a follow-up survey.

Declaration wording

When the supplier letters are on file and nothing is declarable:

No substances on the current REACH SVHC Candidate List (ECHA, as of [date]) are present above 0.1% w/w in the delivered articles, per supplier declarations on file.

When a part does contain one, for example 12L14:

The delivered articles contain lead (CAS 7439-92-1), an SVHC on the REACH Candidate List, at approximately 0.15 to 0.35% w/w as an alloying element of AISI 12L14 steel. No other SVHC is present above 0.1% w/w.

YEARLY

The Candidate List changes twice a year, so supplier letters age. Ask TYRO and each finisher for a fresh letter every January so nothing on file is older than a year.

DFARS in detail

Defense Federal Acquisition Regulation Supplement. The Department of Defense's add-on to federal purchasing rules. Its clauses flow down every PO in a defense supply chain until they reach a job shop. When a customer writes "DFARS compliant material" they almost always mean one clause: **252.225-7009, Restriction on Acquisition of Certain Articles Containing Specialty Metals.**

The rule: any specialty metal in a part delivered to DoD must be **melted or produced in the United States or a qualifying country**. Qualifying countries are mostly NATO allies plus Japan, Australia, Israel and a few others. **Vietnam, Thailand, China, India, Korea, Taiwan, Mexico and Brazil are not qualifying countries.**

What counts as a specialty metal (defined by chemistry, not by name)

- **Steel** containing more than 0.25% of chromium, nickel, molybdenum, titanium, vanadium, aluminum, cobalt, niobium or tungsten; or manganese over 1.65%, silicon over 0.60%, or copper over 0.60%.
- **Nickel, iron-nickel and cobalt alloys** with more than 10% other alloying metals.
- **Titanium** and titanium alloys. **Zirconium** and zirconium alloys.

Material	Specialty metal?	Melt restriction applies?	Sourcing action
A1008 / A36 plain carbon steel	No	No	None. Foreign mill certs (Vietnam, Thailand) are acceptable.
Galvanized / galvaneal	No (base steel is plain)	No	None.
5052 / 6061 aluminum	No (aluminum is not on the list)	No	None. Customer may still ask for domestic under Buy American; read the PO.
304 / 316 stainless	YES (Cr, Ni)	YES	Write "DFARS 252.225-7009, domestic or qualifying-country melt" on the PO to TYRO. Cert must show melt country.
4130 and other alloy steels	YES	YES	Same as stainless.
Titanium	YES	YES	Same as stainless.

Exceptions we use

- **Commercial fasteners.** PEM inserts, screws and rivets are exempt if the maker certifies it buys at least 50% of its specialty metal from domestic or qualifying sources. PEM and most fastener houses publish this. Keep it on file.
- **De minimis.** Non-compliant specialty metal is allowed if it is under 2% by weight of all specialty metal in the delivered item. Saves a job when one small stainless bushing turns out to be foreign.
- **COTS.** Commercially available off-the-shelf items are exempt. Does not cover parts made to a customer drawing.

Other DFARS clauses that can land on a PO to us

Clause	What it is	What it means for us
252.204-7012 and CMMC	Cybersecurity. Safeguarding covered defense information; NIST SP 800-171 controls.	A much bigger lift than any material rule. Confirm scope BEFORE accepting the PO.
252.246-7007 / 7008	Counterfeit electronic parts.	Buy electronics only from authorized sources with traceability. Rare for sheet metal; appears on assembly work.
252.225-7052	Magnets and tungsten from China, Russia, North Korea, Iran.	Only if we supply assemblies containing magnets or tungsten.
252.225-7001	Buy American.	Domestic end product; component cost test. Read the PO for which version applies.

COC WORDING

Specialty metals comply with DFARS 252.225-7009. Carbon steel and aluminum are not specialty metals as defined in the clause.

The second sentence stops a customer bouncing a Vietnamese or Thai carbon steel cert back to us.

Shop procedure

At order entry

- Read the PO, drawing notes and any attached quality clauses for RoHS, REACH, DFARS, 2011/65/EU, 2015/863, 1907/2006, 252.225-7009, or "hazardous substance declaration".
- Check the customer's supplier terms we signed at onboarding. Electronics and medical customers often put the flowdown there rather than on each PO.
- Flag the order in Orders with the applicable rule(s) so purchasing and QC both see it.

At purchasing

- **RoHS / REACH job:** order bare sheet normally. Confirm the distributor letter on file covers the alloy. Specify trivalent on every finish PO with "state RoHS compliance on cert." Check hardware finish (no yellow zinc).
- **DFARS job:** for stainless, alloy steel or titanium write "DFARS 252.225-7009 domestic or qualifying-country melt" on the PO to the distributor. Carbon steel and aluminum need nothing extra.
- **Leaded alloys** on a RoHS or REACH job: stop and confirm with the customer, or substitute 1215 / 6061 / lead-free brass.

At receiving and QC

- Match heat number on the cert to the tag on the material and to the PO.
- RoHS: look for the statement on the cert. If absent, attach the distributor letter. Check for a page 2.
- DFARS: for specialty metals, confirm the mill and country on the cert are USA or qualifying.
- Finish: finisher cert must state trivalent / RoHS. Reject the cert, not the parts, if the wording is missing; ask the finisher to reissue.

At shipping (Certificate of Conformance)

- Cite each rule by its full name and attach the supporting certs and letters. Never make a bare claim without a document behind it.
- If something is unproven, say so: "material meets ASTM A1008 CS Type B; RoHS declaration pending from supplier."

Quick reference: what to write on the CoC

Rule	Wording
RoHS	Materials and finishes are RoHS compliant per Directive 2011/65/EU as amended by (EU) 2015/863, per mill certificate [no.], heat [no.], and finisher certificate [no.].
REACH	No substances on the current REACH SVHC Candidate List are present above 0.1% w/w in the delivered articles, per supplier declarations on file. [Or name the substance and CAS number.]
DFARS	Specialty metals comply with DFARS 252.225-7009; melt country per attached mill certificate. Carbon steel and aluminum are not specialty metals as defined in the clause.

Documents to keep on file (refresh every January)

Document	From	Covers	On file?	Date
RoHS / REACH statement, carbon steel (A1008, A36, A653)	TYRO	RoHS, REACH	☐	
RoHS / REACH statement, aluminum (5052, 6061)	TYRO	RoHS, REACH	☐	
RoHS / REACH statement, stainless (304, 316)	TYRO	RoHS, REACH	☐	
Standing instruction: DFARS melt on all stainless / alloy for flagged POs	TYRO	DFARS	☐	
RoHS / REACH statement + list of trivalent bath names	Chem film / plating house	RoHS, REACH	☐	
RoHS / REACH statement	Anodize house	RoHS, REACH	☐	
RoHS / REACH statement	Powder coat house	RoHS, REACH	☐	
RoHS statement + DFARS fastener certification	PEM / fastener distributor	RoHS, REACH, DFARS	☐	
Customer supplier-terms review (which rules each customer flows down)	Internal	All	☐	

Worked examples from our own certs (Sept 2026, TYRO PO 26404)

- **20 ga A1008 CS-B, CSVC (Vietnam), cert 241025F00033, heat 4921622:** Specification block reads “ASTM A1008 CS TYPE B / ROHS COMPLIANT.” RoHS proven by the cert itself. Not a specialty metal, so the Vietnamese melt is fine for DFARS.
- **16 ga A1008 CS-B, TCRSS (Thailand), cert 15444, heat K796613:** no RoHS line, no Pb or Cd columns, cert says page 1 of 2 and page 2 was missing. RoHS unproven until TYRO sends page 2 or a letter. Not a specialty metal, so fine for DFARS.

This guide is a working summary for shop use, not legal advice. Directive text, exemption expiry dates, the SVHC list and the DFARS qualifying-country list change; verify against the current source when a customer is strict.